

MICROMOLD · DESIGN PRECIS

MicroMold design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Figures brought into line with MMD-CAL-001 v0.5 - torque-limiting socket and plunger rest in Table 1 (items 19 to 21), overload 8.3 kN with the socket, mass 39.8 kg, press USD 590
0.7	2026-10-02	Amish Chadha	Decisions of 2026-10-02 (MMD-DEC-001 v0.2): lift-out loading with a plunger rest, torque-limiting socket against overload, first product mold, first candidate partner
0.6	2026-10-01	Amish Chadha	Constructable design (MMD-DDR-003) and build plan (MMD-BLD-001); budget read as a value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish (\$520)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update from MMD-CAL-001 and MMD-DDR-001 (ratchet handle, 4 mm nozzle, load cell spacer, 20 mm bracket, nozzle zone shield, side fume

			hood, raised nozzle; numbers and design choices updated)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, design choices, safety, open questions, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

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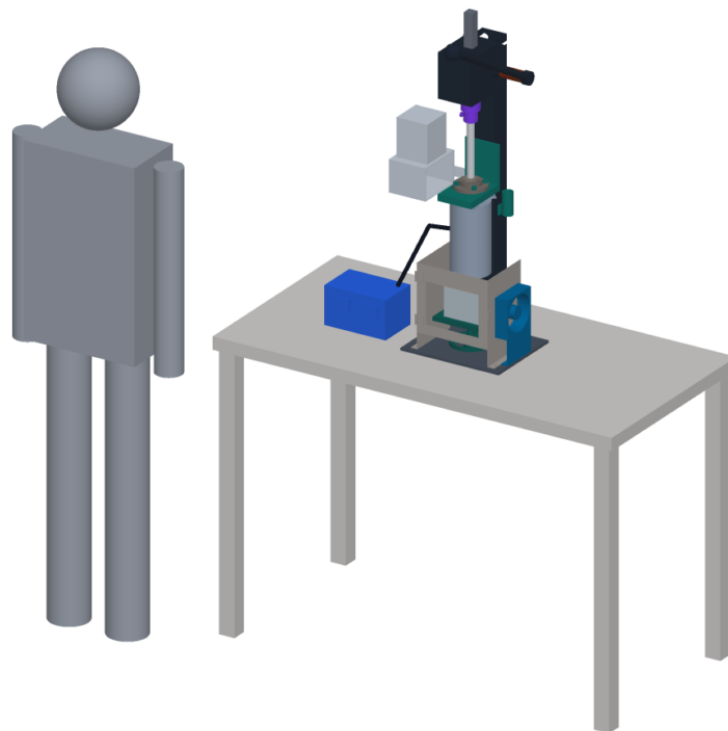
■ Summary

MicroMold is a bench-top, hand-operated plunger injection press for recycled HDPE, PP, LDPE and PS. The rack-and-pinion head of a 1 t arbor press, mounted on a taller steel column and turned by a 450 mm ratchet handle through a torque-limiting socket set to 180 N m, drives a 22 mm plunger down a vertical steel barrel heated by two 300 W band heaters. Melt leaves a heated 4 mm nozzle into a two-plate aluminum mold that a screw lift table holds against the nozzle, inside a perforated nozzle zone shield, where a small fan cools the mold, and a side hood with a duct fan draws fumes from the funnel. The TRL 3 calculations (MMD-CAL-001 v0.5) give a shot of up to 34.2 g at 8.9 MPa (89 bar) with 250 N on the handle, delivered in about four ratchet pulls; a warm-up of 12.3 min; and about 11.6 parts per hour from 753 W of single-phase power. Value-engineering target: USD 520. Estimated cost of the constructable design: USD 590 (USD 70 over the target) for the press, and USD 684 with one mold, which is tooling. It weighs about 39.8 kg against a 40 kg target. All figures are estimates. The design was made buildable on 2026-10-01 (MMD-DDR-003) and the prototype build plan is MMD-BLD-001 (docs/05-build-plan.md).

MicroMold

github.com/BoujeeEnjinia1701/micromold

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: workbench (0.9 m), 1.75 m person for scale

Figure 1. MicroMold on a 0.9 m workbench with a 1.75 m person for scale. Plunger raised for loading; test mold seated on the nozzle inside the shield. Generated from the parametric model `cad/src/model.py`.

■ How it works

1. **Heat.** Two PID controllers drive the barrel zone (two 300 W band heaters) and the nozzle zone (a 100 W band heater) to the set point for the resin, for example 210 °C for HDPE. An insulation jacket and a perforated guard keep the outer surface near 48 °C. The duct fan runs whenever the heaters are on. Feedstock is flake from injection-molded items such as caps, crates and buckets; bottle-grade HDPE is too stiff for a hand press.
2. **Load.** The operator pulls the ball-lock pin, lifts the plunger out and sets it on a plunger rest on the column, never on the bench (decided by Amish, 2026-10-02), pours washed, dried flake into the funnel, refits the plunger and tamps each of two or three top-ups down with the press itself, because loose flake has about a third to a half of the density of melt and an untamped charge wastes stroke (MMD-CAL-001, A4).
3. **Soak.** The fresh charge melts above the melt left from the last shot. The barrel holds 2.2 shots, so each charge soaks for about two cycles, about 12 min, against the 9.3 min a 22 mm column of tamped flake needs.
4. **Clamp.** The operator bolts the two mold plates together, sets the mold on the lift table, closes the shield's front and turns the handwheel nut, reached under the front, until the nozzle seats in the mold's spherical seat.
5. **Inject and hold.** The operator fills the mold in about four pulls of the ratchet handle, each moving the plunger about 31 mm, over about 10 s; the pawl holds the ram between pulls. The operator then holds pressure for about 30 s while the gate freezes. A load cell above the plunger shows its force, so the melt pressure (force divided by 380 mm² of plunger area) can be read and logged.
6. **Cool and eject.** With the mold cooling fan running, the plaque cools in the mold for about 1 min, then the mold is lowered, unbolted and opened. Sprue and purge go back to the regrind bin.

MicroMold: material flow per shot, grams (estimates, 30 g HDPE test part)

github.com/BoujeeEnjinia1701/micromold

CONCEPT, NOT FOR FABRICATION

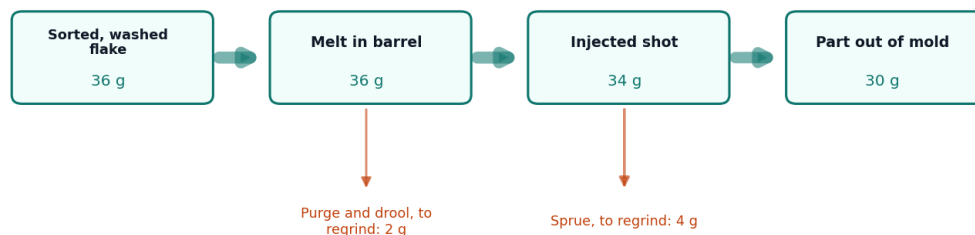


Figure 2. Material flow per shot for the 30 g reference part. All values are estimates: about 2 g of purge and drool and 4 g of sprue return to regrind.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Choice	Notes
1	Base plate	320 x 260 x 8 mm steel, bolted to the bench	Carries the column and the mold clamp; 8 mm from MMD-DDR-003
2	Column and drive head	Rack-and-pinion head of a 1 t arbor press on an 80 x 60 x 3 mm steel tube column 0.9 m tall; 450 mm, 1/2 in drive ratchet handle on the pinion shaft	Ratchet added at TRL 3: one pull moves the plunger only about 31 mm
3	Rack ram	From the arbor press, 28 mm square, 255 mm or longer	150 mm of travel with the rack still engaged; to confirm on the chosen press
4	Plunger load cell	10 kN compression cell with HX711 amplifier and display, on a 10 mm G-11 glass-epoxy spacer	The spacer keeps the cell near 27 °C instead of about 94 °C
5	Plunger	22 mm ground steel rod, 200 mm, cross-drilled for a pinned floating coupling	0.05 to 0.10 mm diametral clearance (22 H8/f7); no seal
6	Heated barrel	42 mm steel bar bored and honed to 22 mm, 260 mm long, 80 mm flange and loading funnel	Machine shop part
7	Band heaters	Two 300 W mica band heaters, 42 mm ID x 50 mm	Barrel zone, one PID loop; 4.5 W/cm ² ; 300 W from MMD-DDR-002
8	Nozzle and nozzle heater	Steel nozzle with a 4 mm orifice, radiused tip; 100 W band heater; tip at 190 mm above the bench	Second PID loop; orifice enlarged from 3 mm at TRL 3
9	Barrel shelf and heat break	20 mm steel shelf and 8 mm back plate welded to the column; the 100 mm flange sits on four mica pads, held by four M8 cap screws	Carries the plunger reaction into the column; back plate from MMD-DDR-003
10	Insulation jacket and guard	25 mm mineral wool in an aluminum skin, perforated steel guard	Skin about 48 °C
11	Mold clamp	170 x 126 mm lift table on a Tr20 x 4 screw, raised by a handwheel nut on the base plate; the screw drops through a hole in the bench	Table top 60 to 160 mm: stacks of 30 to 120 mm; screw lift from MMD-DDR-003
12	Aluminum mold set	Two 6061 plates, 120 x 90 x 45 mm each, four M10 x 70 cap screws in thread inserts, two dowels, sprue tapering 5 to 7 mm	First mold: test plaque; tooling outside the press budget
13	Control box	Two PID controllers with set point limit, two 25 A SSRs on a heat sink, fused IEC inlet, double-pole switch, independent thermal cut-out, fan outlet, force display	Mains enclosure; earthed
14	Heater and sensor wiring	High-temperature wire in glass-fiber sleeving, two K-type thermocouples	250 °C rated at the barrel
16	Nozzle zone shield	Perforated steel sides and hinged front around the nozzle and mold	Added at TRL 3 for R11 and melt spit
17	Fume hood and duct fan	Side hood, 150 x 100 mm face, 100 mm from the funnel axis; 100 mm inline fan; 3 m of duct	Added at TRL 3 (MMD-DDR-001 D9); about 124 m ³ /h
18	Mold cooling fan		Added by MMD-DDR-002; mold about 47 °C instead of about 92 °C

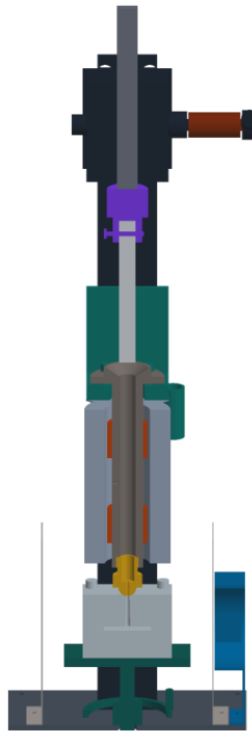
#	Component	Choice	Notes
		120 mm mains axial fan, about 18 W, on an angle bracket on the base plate, blowing on the mold through the shield side	
19	Construction parts	Head mounting plate, plunger coupling and ball-lock pin, ratchet adapter, hood arm	Added to make the design buildable (MMD-DDR-003)
20	Plunger rest	A cup on a plate on two M6 rivnuts in the column's right wall, 400 mm above the bench	Holds the plunger upright, tip down, while the barrel is loaded; the hot plunger is never laid on the bench (decision of 2026-10-02)
21	Torque-limiting socket	1/2 in drive, set to 180 N m, between the ratchet adapter and the ratchet	Slips at about a 400 N pull so the ram cannot exceed the press rating; checked against the load cell at TRL 4 (decision of 2026-10-02)

Item 15 (hardware and consumables) is in the BOM but not modeled. The general arrangement is drawing MMD-DWG-001 (cad/drawings/).

MicroMold: cutaway

github.com/BoujeeEnjinia1701/micromold

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 3. Section on the injection axis: drive head and ram at the top, load cell and spacer (purple), plunger raised above the funnel, barrel with the two band heaters inside the jacket and guard, bracket (teal), nozzle (gold) seated on the mold with its sprue and plaque cavity, lift table and screw jack on the base plate, and the sides of the nozzle zone shield.

■ Numbers from MMD-CAL-001

All values are first-principles estimates. Assumptions are in MMD-CAL-001, Table 1; the tags point to the lines of `docs/04-calcs/sizing.py` that print them.

Table 2. Key numbers.

Quantity	Value	Requirement
Swept volume and shot	45.6 cm ³ , 34.2 g HDPE ideal; about 25 g if the fresh charge is not tamped [A1, A2, A4]	R2 at risk
Drive ratio and efficiency	22.5:1; 0.60 design, 0.84 calculated [B1]	
Plunger force and melt pressure at 250 N	3.38 kN and 8.9 MPa (89 bar, 1,288 psi) at 0.60; 12.4 MPa at 0.84 [B2]	R3 met on paper, 11 % margin
Stroke per ratchet pull	31.4 mm per 90° pull; 3.8 pulls per shot [B5]	
Overload (700 N on the handle)	13.2 kN, 34.7 MPa, 1.35 times the press rating without a limiter; with the torque-limiting socket at 180 N m (198 N m worst case) 8.3 kN, 21.8 MPa, 0.85 times the rating [B4, B4a, B4b]	Parts sized for the limited case
Fill pressure, 4 mm nozzle, 10 s fill	About 4.8 MPa [E3]	
Mold clamp limit	45 cm ² projected area at 8.9 MPa [D2]	R7 met on paper
Warm-up to 220 °C	12.3 min barrel zone with 2 x 300 W, 6.6 min nozzle [G4]	R5 met on paper
Standing losses at 220 °C	103 W; jacket skin 48 °C [G2, G3]	R11 met on paper
Melt soak for a 22 mm column	9.3 min for tamped flake [H1]	
Cycle	5.2 min, 11.6 parts per hour with the mold fan (soak limited); mold about 47 °C [H3, H6]	R6 met on paper
Electrical load	753 W; 3.3 A at 230 V, 6.3 A at 120 V [I1]	R10 met on paper
Energy per shot	About 20 Wh, 0.66 kWh per kg of parts [I2]	
Fume hood flow	About 124 m ³ /h [J1]	R13 met on paper
Mass	39.8 kg without the control box, against 40 kg [K1]	R9 at risk
Size	320 x 260 mm base; handle 1,094 mm above the bench at the start of a pull [K2]	R9 met on size
Parts cost	\$590 press, against a \$520 value-engineering target; \$684 with one mold [L1, L2]	R14 USD 70 over the value-engineering target

■ Key design choices

These follow MMD-DDR-001 and MMD-DDR-002. Each was decided by Amish, 2026-09-25: go with recommendation.

- **Drive: rack and pinion from an arbor press (D1).** The rack, pinion and head of a 1 t arbor press on a taller column give 22.5:1 from one bought part. A screw press stays as a variant for small, thick parts. At TRL 3 the fixed handle is replaced by a ratchet handle, because a shot needs 344° of pinion rotation.
- **Vertical barrel, mold below (D2).** Gravity keeps the melt at the nozzle and the operator loads from the top. The mold is changed at bench height.
- **Plunger, not screw (D3).** A reciprocating screw would mix and melt better but needs a motor, gearbox and a much more complex barrel.
- **Two heat zones (D4).** A separate nozzle zone stops the nozzle freezing between shots at the cost of a second PID and SSR.
- **Bolted two-plate aluminum molds on a lift table (D5).** Bolts are slow but cheap and hold 45 cm² at full pressure. A toggle clamp frame (about \$40) is a later upgrade. The nozzle tip is now 190 mm above the bench so that 120 mm stacks fit.
- **Load cell for pressure indication (D6).** A number for each shot helps a group learn and record settings. A glass-epoxy spacer keeps the cell cool.
- **First mold: a test plaque (D7).** A 64 x 50 x 6 mm plaque shows fill, shrinkage and surface quality and can be cut into test bars. The first product mold is a flat, single-cavity part of 25 g or less that fits the 120 x 90 mm plates, such as a coaster, tile or cable clip, chosen with the partner from what it can sell (decided by Amish, 2026-10-02).
- **Budget (D8, DDR-002).** The budget covers the press and is \$520, approved by Amish on 2026-09-26 to cover the priced BOM; molds are tooling, reported separately (R8). On 2026-10-01 Amish set out that the budget is a hypothetical value-engineering target, not a limit. The constructable press is estimated at \$544, \$24 over the target (MMD-DDR-003).
- **Fume control (D9).** A side hood with a duct fan at the funnel, plus a written condition to run only under it or outdoors.
- **Heaters, mold cooling, mass and feedstock (DDR-002).** Two 300 W barrel bands for a 12.3 min warm-up; a 120 mm fan at the mold so parts eject from a mold near 47 °C; a 40 kg mass target for a bench-bolted press; and injection-grade flake only.
- **Co-design partner type (D10).** The first partner is to be a group that already shreds HDPE or PP; the first candidate to approach is a Precious Plastic workspace near Irving, Texas (decided by Amish, 2026-10-02).

MicroMold: exploded viewgithub.com/BoujeeEnjinia1701/micromold

CONCEPT, NOT FOR FABRICATION

- 1 Base plate
- 2 Column, drive head and ratchet handle
- 3 Rack ram
- 4 Plunger load cell and spacer
- 5 Plunger, 22 mm
- 6 Heated barrel
- 7 Band heaters, 2 x 300 W
- 8 Nozzle and nozzle heater
- 9 Barrel bracket and heat break
- 10 Insulation jacket and guard
- 11 Mold clamp (screw lift table)
- 12 Aluminum mold set
- 13 Control box with PID and SSR
- 14 Heater and sensor wiring
- 16 Nozzle zone shield
- 17 Fume hood and duct
- 18 Mold cooling fan
- 20 Plunger rest
- 21 Torque-limiting socket, 180 N m



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 4. Exploded view with BOM numbers.

■ Safety

Safety: MicroMold combines mains voltage, surfaces and molten plastic at up to 260 °C, stored pressure in the melt, a long ratchet lever and plastic fumes. It is a concept for a supervised workshop, not a consumer appliance.

- **Hot plunger.** The plunger's top end reaches about 94 °C. Lift it out for loading only in heat-resistant gloves and set it on the plunger rest on the column, never on the bench.
- **Hot surfaces and melt.** The barrel, nozzle, mold and purge reach 180 to 260 °C, and molten plastic sticks to skin. The funnel top is a working surface at about 150 °C. Keep the jacket guard and the nozzle zone shield in place, wear heat-resistant gloves, long sleeves and eye protection, and never look down the barrel.
- **Pressure.** Trapped melt can spit from the nozzle or the parting line. Without a limiter, an operator hanging body weight on the handle could reach about 35 MPa, which would open the parting line of the test mold (MMD-CAL-001, D3); with the socket the most is about 22 MPa. Close the shield before injecting and keep faces away from the axis. A torque-limiting socket on the ratchet adapter, set so the ram cannot exceed the press rating, limits the force whoever pulls (decided by Amish, 2026-10-02); it is checked against the load cell at TRL 4.

- **Fumes.** Process only HDPE, PP, LDPE and PS within their published ranges. Never heat PVC, which releases hydrogen chloride, or unknown and mixed plastics ([Precious Plastic Academy](#)). Overheated PS can release styrene. Run the hood fan whenever the heaters are on, or work outdoors. Set point limits of 260 °C are set in both controllers.
- **Mains electricity.** Heaters run at mains voltage next to steel parts. Earth the frame and every metal part, use a fused inlet, double-pole switch and an RCD (30 mA) or GFCI supply, rate wiring at the barrel for 250 °C, and fit an independent thermal cut-out. Mains wiring must be done or checked by a qualified electrician to local electrical code.
- **Moving parts and lever.** The ratchet holds the ram under load: release the pawl only with the handle in hand. The ram and pinion can pinch; keep hands clear of the rack. The mold cooling fan runs at mains voltage: keep its finger guard fitted and its lead clear of the hot barrel and nozzle.
- **Fire.** Unattended heaters are a fire risk. Switch off at the inlet after use; keep flake and rags away from the barrel.
- **Product use.** Recycled parts of unknown history are not suitable for food contact, toys for young children, medical or safety-critical uses.

■ Open questions after TRL 3

- Ram length and travel of real 1 t arbor presses: is a 255 mm ram with 150 mm of engaged travel available, and how heavy is the head? This sets R9, now at 39.8 kg against 40 kg.
- How dense is a tamped, part-melted charge at injection time? This sets R2.
- Is the fan enough to keep the mold near 47 °C in practice, and how dense is the charge after the shorter soak-limited cycle? This sets R6.
- Nozzle temperature for each resin, and mold design rules for the group's own machinists: draft, venting, gate size and ejection.
- Which first products would sell locally? The first product mold is a flat part of 25 g or less, such as a coaster, tile or cable clip (decided 2026-10-02); which one is chosen with the co-design partner.

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [MMD-DWG-001](#).